

Answer: A

- 7 Let A = area of cross-section of ice, L = length of ice, h = length of ice submerged in water, ρ_w = density of water and ρ_i = density of ice

Upthrust on ice when it is fully submerged in water = $V\rho g = AL\rho_w g$

Upthrust on ice when it is floating in water = Weight of ice

$$Ah\rho_w g = AL\rho_i g$$

$$h = \frac{L\rho_i}{\rho_w} = \frac{0.9L}{1.0}$$

Upthrust when fully submerged / Upthrust when floating

$$= \frac{AL\rho_w g}{Ah\rho_w g} = \frac{L}{h} = \frac{L(1.0)}{0.9L} = 1.1$$

Answer: D

8 $D = E \times (1 \text{ m}^2) \times 1 \text{ m} = 1 \text{ m}^3$